

AI FOR AMERICANS FIRST

AI Protectionism, Energy and Semiconductors:
US/Europe Divergence Trajectories 2024–2030

Integrated Geostrategic and Economic Analysis

Chapter VI

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77% global AI compute = USA **1.6×** energy cost EU/US **3.4:1** CACI
ratio US/EU (Power Mode)

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7 chapters • 4 prospective scenarios • 3 geographic zones

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CHAPTER VI

Consequences for France and Europe

The preceding chapters established the diagnosis (III), the mechanisms (IV), and the possible trajectories (V). This chapter details the concrete consequences for French and European actors, distinguishing three levels of analysis: sectoral breakdown (which sectors are most exposed?), differentiation by actor type (large corporations, SMEs, startups, public sector), and second-order effects (brain drain, R&D offshoring, regulatory fragmentation). The analysis primarily relies on Scenario A (the most probable) and Scenario B (the most severe), while noting the bifurcations specific to Scenarios C and D.

6.1 Sectoral Analysis: Differentiated Exposure to the Compute Asymmetry

6.1.1 Financial Services: Advanced Dependence

The French financial sector is the most advanced in AI adoption and, paradoxically, the most exposed to geopolitical vendor lock-in risk. BNP Paribas, Société Générale, Crédit Agricole, and AXA are massively deploying AI solutions for fraud detection, credit scoring, algorithmic trading, and risk optimization. AXA and BNP Paribas are among Mistral AI's first enterprise clients in France.¹ These deployments largely rely on US cloud infrastructure (AWS for BNP Paribas, Azure for Société Générale). Productivity gains observed in the global financial sector are among the highest (the IMF cites microeconomic gains of 20 to 40% on compliance and analysis tasks), as it is a sector with high cognitive intensity and high wages — two factors that maximize automation's return on investment.²

The risk specific to the financial sector is twofold. On one hand, European regulations (DORA, AI Act, GDPR) impose localization and auditability requirements that create tension with US cloud dependence: French banks must guarantee that their clients' data is not accessible to American authorities (CLOUD Act), while relying on AWS and Azure for computing power. On the other hand, under Scenario B, increased costs for cutting-edge AI cloud access would directly hit the most compute-intensive applications (risk models, Monte Carlo simulations, training of specialized LLMs). The productivity gap with American banks (JPMorgan, Goldman Sachs, each investing several billion per year in AI) would widen by 3 to 5 additional percentage points per year.

6.1.2 Automotive and Aerospace: Compute-Intensive and Vulnerable

The European automotive industry has already experienced a revealing precedent: the 2022 semiconductor shortage cost the EU auto sector alone approximately €100 billion (Chapter IV). AI is transforming this sector along three axes: autonomous driving (training perception models requiring tens of thousands of GPUs), production optimization (digital twins, predictive maintenance), and assisted design (aerodynamic simulation, virtual crash tests). Stellantis (formerly PSA) signed a €100 million partnership with Mistral AI to integrate AI across all its operations, from transport to logistics.³

Aerospace (Airbus, Safran, Thales, Dassault Aviation) presents a similar profile but aggravated by the defense dimension. Complex aerodynamic simulations, fleet predictive maintenance, and autonomous weapons system design are extremely compute-intensive applications. Airbus relies on AWS and Azure for its cloud workloads. Under Scenario B, restrictions on advanced GPUs would directly affect simulation capabilities; under Scenario D, the pressure to repatriate workloads to sovereign infrastructure would create considerable transition costs but would reduce strategic vulnerability.

6.1.3 Healthcare and Life Sciences: Data Sovereignty at Stake

The healthcare sector presents a different vulnerability: compute intensity is lower (except for AI-driven drug discovery, which requires massive GPU clusters), but data sensitivity is at its maximum. The European Health Data Space (EHDS) strictly regulates the processing of medical data. Sanofi, one of the world's leading pharmaceutical groups, has invested \$1 billion in AI partnerships (including OpenAI and Owkin, a French startup specializing in AI for clinical research).⁴ Novo Nordisk, though Danish, illustrates the European challenge: over \$8.2 billion in R&D in 2024, with growing reliance on AI for digital twins of the human body and accelerating drug discovery.⁵

The impact of protectionism scenarios here is indirect but structural. If access to cutting-edge GPUs is restricted, AI drug discovery projects (which require model training over several weeks on thousands of GPUs) will be slowed or relocated to the United States. Owkin, founded in Paris, has already opened offices in New York and could transfer its most compute-intensive workloads there if European costs became prohibitive.

6.1.4 Robotics and Manufacturing: The Energy Factor

AI robotics and industrial automation add an additional energy dimension. Training perception and control models for industrial robots is compute-intensive, but it is mainly the large-scale deployment of AI-equipped robots (edge computing, embedded inference) that multiplies industrial energy demand. The convergence of data centers + edge compute + robots could add 20 to 30% to industrial energy demand by 2030 (Chapter III), an estimate still poorly quantified but recognized as a critical sensitivity variable.

France has significant players: Exotec (logistics robotics, valued at over €2 billion), Wandercraft (exoskeletons), Aldebaran/SoftBank Robotics (humanoid robots, founded in France). These companies depend on AI hardware (Nvidia GPUs, specialized chips) to train and deploy their systems. Under Scenario B, GPU quotas would directly affect the pace of innovation; under Scenarios C and D, investment in European Gigafactories would provide the necessary compute, but with a 2 to 3 year lag behind American competitors.

Table 12. French sectoral exposure to the AI compute asymmetry. Source: author.

Sector	Compute Intensity	Data Sensitivity	US Cloud Depend.	Scenario B Risk
Finance	High	Very high	70–80%	Lock-in + surcharges: -3 to -5 pts productivity/yr
Auto/Aero	Very high	High	60–70%	Slowed simulations, R&D offshoring

Health/Pharma	High (drug disc.)	Maximum	40–60%	Drug discovery relocated to US
Robotics/Manuf.	High	Moderate	50–65%	Slowed innovation + energy surcharge
Defense/Space	Very high	Critical	Variable	Maximum strategic vulnerability

6.2 Differentiation by Actor Type

6.2.1 Large Corporations: Constrained Beneficiaries

Large French companies (CAC 40 and SBF 120) are the primary short-term AI beneficiaries and the most exposed to medium-term dependence. They have the budgets to access US cloud and the teams to deploy AI, but this adoption reinforces vendor lock-in with each iteration. The migration cost (switching cost) of a complete cloud ecosystem (AWS → OVHcloud, for example) is estimated at 12–18 months of development and millions of euros in re-architecture, making it economically irrational absent strong regulatory constraint. Under Scenario A, they continue adopting AI via US cloud, accumulating growing dependence while benefiting from real productivity gains. Under Scenario B, the sudden increase in compute costs presents them with a dilemma: absorb the surcharges (margin compression) or slow AI projects (competitiveness loss).

6.2.2 SMEs and Mid-Cap Industrial Firms: Progressive Exclusion

SMEs and mid-cap companies represent the French industrial fabric (4,000 mid-caps, 140,000 SMEs). Their access to cutting-edge AI is already constrained by costs: training a specialized model costs several hundred thousand euros, beyond the reach of most SMEs without subsidies. McKinsey (December 2025) observes that AI productivity gains are concentrated in large companies, creating an "intra-European productivity divide" between adopters and non-adopters.⁶ Under Scenario B, rising compute costs widen this divide: SMEs forgo cutting-edge AI and opt for degraded solutions (lightweight open-source models, local inference on limited hardware), progressively losing competitiveness against American SMEs that benefit from domestic tariff-exempt compute.

The EuroHPC AI Factories, designed to prioritize access for startups and SMEs, could mitigate this effect (Scenarios C and D). But their total capacity (~475,000 GPUs, on the order of a single US hyperscaler) is insufficient to serve the entire European economic fabric. The gap between public compute supply and market demand is structurally deficit.⁷

6.2.3 French AI Startups: Between Championship and Dependence

The French AI startup ecosystem is the most dynamic in Europe. Mistral AI, valued at €11.7 billion (Series C of September 2025, led by ASML), is Europe's generative AI champion.⁸ The company has raised €2.8 billion in total, employs approximately 700 people, and is developing Mistral Compute, a sovereign cloud platform with 18,000 Nvidia Grace Blackwell GPUs deployed in a 40 MW data center in Essonne, powered by French nuclear energy.⁹ In February 2026, Mistral announced a €1.2 billion investment for a data center in Sweden (Borlänge), and the acquisition of startup Koyeb to strengthen Mistral Compute.¹⁰

The Mistral case illustrates both the potential and limits of European AI sovereignty. On the potential side: the company proves that a European startup can reach global scale, attract

massive investments (ASML, Nvidia, Bpifrance, a16z), and build its own compute infrastructure. On the limitations side: Mistral remains dependent on Nvidia GPUs (no European alternative for AI accelerators), initially used Microsoft Azure and Google Cloud for training, and its investor Microsoft holds a strategic position (€15 million investment conversion). The company illustrates application-level sovereignty (models, platforms, services) without hardware-level sovereignty — precisely the "junior partner" position described in Scenario C.

Beyond Mistral, the French ecosystem includes Hugging Face (valued at \$4.5 billion, model hosting platform, headquartered in Paris but with US infrastructure), LightOn (specialized enterprise models), H Company (formerly Holistic AI), Owkin (AI for healthcare), and Scaleway (French sovereign cloud, first European access to Nvidia Blackwell GPUs). AI funding in Europe grew by 55% in Q1 2025, with 12 new unicorns in H1. France has been Europe's top destination for foreign AI investment for the fifth consecutive year.¹¹

6.2.4 Public Sector and Defense: The Sovereignty Imperative

The French public sector represents a special case. The Ministry of the Armed Forces launched GenIAI.intradef in February 2025, a secure conversational agent based on Mistral AI.¹² The Direction Générale de l'Armement (DGA) and ANSSI are working on classification frameworks for sovereign AI. For defense and intelligence, dependence on US cloud is unacceptable regardless of the scenario: classified workloads must be executed on national infrastructure. This captive market provides a guaranteed demand base for sovereign cloud providers (OVHcloud S3ns, Thales S3ns, Scaleway), but its size remains limited compared to the commercial market.

6.3 Second-Order Effects

6.3.1 Brain Drain and AI Talent Capture

The compute asymmetry produces a talent attraction effect toward the United States. European AI researchers and engineers are drawn by: (i) American compensation packages (salary + equity), which are structurally superior to European standards; (ii) access to cutting-edge compute, a necessary condition for frontier research; (iii) the ecosystem (proximity to Big Tech, VCs, research community). Martens (Bruegel) notes that personnel costs (salaries + equity) are often the most significant component of AI model development costs, and that US companies compete fiercely to recruit the best global talent.¹³

This brain drain directly weakens the European CACI's L(r) factor. It is partially offset by the excellence of French training institutions (ENS, École Polytechnique, INRIA, Parisian universities), which produce a continuous flow of talent — but a significant proportion departs for the United States after their doctorate. Mistral AI, founded by DeepMind and Meta alumni, illustrates the possibility of retaining (or repatriating) talent, but it is the exception rather than the rule. Under Scenarios C and D, the deployment of Gigafactories and Mistral Compute could create a sufficiently attractive ecosystem to retain more talent.

6.3.2 R&D Offshoring

The compute asymmetry produces pressure to offshore the most compute-intensive R&D activities. This phenomenon is already observable: the largest AI R&D centers of European

companies are often located in the United States (Bosch AI in Sunnyvale, SAP AI in Palo Alto, DeepMind in London but owned by Google). Under Scenario B, this pressure intensifies: 15 to 20% of critical AI projects could be relocated to the United States (notably near Nvidia/TSMC fabs in Arizona, or Virginia data centers). Under Scenarios C and D, offshoring is mitigated by local compute availability, but does not disappear completely because the FLOP cost gap remains positive in favor of the US in all scenarios.

6.3.3 Regulatory Fragmentation: The AI Act as a Double-Edged Sword

The European AI Act, progressively entering into force since 2024, produces ambivalent effects in the context of American protectionism. On one side, it creates additional compliance costs for European companies (model transparency, copyright management in training data, security audits). Draghi (2024) observed that European data restrictions create high costs and hinder model training. Martens (Bruegel, 2025) notes that the AI Omnibus risks extending regulatory uncertainty for at least two more years.¹⁴

On the other side, the AI Act creates a barrier to entry for American competitors — an unintentional protectionist effect that could favor European actors compliant by design (Mistral, whose transparent and auditable models are naturally aligned with the AI Act). The Code of Practice on copyright and training data, published in July 2025, imposes constraints that closed models (OpenAI, Anthropic) have more difficulty meeting. Under Scenarios B and D, where geopolitical mistrust is maximal, the AI Act could become a de facto European preference tool, analogous to the GDPR which fostered the emergence of European solutions in personal data processing.

6.4 The France AI Ecosystem: Unique Strengths and Structural Vulnerabilities

The differentiated analysis of the preceding sections allows us to draw up a balance sheet of France's specific strengths and vulnerabilities in the context of AI protectionism.

Table 13. France's strengths and vulnerabilities in the context of AI protectionism. Source: author synthesis.

France's Strengths	France's Vulnerabilities
Nuclear energy (65–70% of mix): competitive cost, low carbon, baseload power for data centers	No AI hardware champion (no European GPU/ASIC design)
AI champion: Mistral AI (€11.7 Bn valuation, Mistral Compute, 18,000 GPUs)	Installed compute: ~5% of global, US/France ratio ~15:1
Training excellence: ENS, École Polytechnique, INRIA, CNRS/IDRIS	Post-doctoral brain drain to US (salaries + equity + compute access)
€109 Bn private AI commitments (February 2025 Summit)	Slow permitting (24+ months), saturated electrical grid
Top EU destination for foreign AI investment (5th consecutive year)	US cloud dependence: 70–80% AI workloads on AWS/Azure/GCP
AI Act: Mistral compliance by design (open-source, transparent, auditable)	AI Act: compliance surcharges, regulatory uncertainty (Omnibus: +2 years)

6.5 Synthesis: France Facing Three Futures

This chapter's analysis converges toward three possible configurations for France by 2030, corresponding to the Chapter V scenario trajectories.

Configuration 1: Dependent Consumer (Scenarios A and B). France adopts AI via US cloud, gains short-term productivity, but accumulates structural dependence placing it in a vulnerable position vis-à-vis any American hardening. Large corporations prosper but are captive; SMEs are progressively excluded from cutting-edge AI; the most promising startups relocate their infrastructure to the United States. Brain drain accelerates. The productivity gap with the United States widens by 5 to 15 cumulative points over five years.

Configuration 2: Energy and Application Hub (Scenario C). France exploits its nuclear advantage to become the energy center of gravity for AI in Europe. Mistral Compute and the Gigafactories provide competitive local compute for inference and fine-tuning. French companies are sovereign in application but dependent on US hardware. The productivity gap is contained to 1–2 points. Brain drain is slowed by the existence of an attractive local ecosystem.

Configuration 3: Pillar of European Sovereignty (Scenario D). American protectionism catalyzes unprecedented mobilization. France, thanks to its nuclear energy, training excellence, and Mistral, becomes the pillar of a European technological sovereignty effort. Massive investment (20 GW dedicated nuclear, RISC-V/DARE, Japan-Korea alliances) creates the conditions for long-term catch-up, but the transition period (2026–2028) is painful. Execution risk is maximum: each year of infrastructure delay extends the vulnerability.

Chapter VII will elaborate the strategic recommendations corresponding to each of these configurations, distinguishing short-term measures (appropriate regardless of scenario) from structural investments (dependent on the chosen trajectory).

Notes

¹ Mistral AI (2025), partnership announcements. AXA and BNP Paribas are among Mistral's first enterprise adopters in France. See NVIDIA Blog (June 2025), "France Bolsters National AI Strategy With NVIDIA Infrastructure".

² IMF (March 2025), op. cit. Financial services are among the sectors with highest AI exposure (Felten and Eloundou indices), with documented microeconomic gains of 20–40% on compliance and risk analysis tasks.

³ Mistral AI / Wikipedia FR (2026). Stellantis (from PSA and Fiat-Chrysler) signed a €100M partnership with Mistral AI in April 2025. CMA CGM also concluded a €100M partnership.

⁴ Sanofi (2024–2025), press releases. Owkin, founded in Paris in 2016, specializes in AI for clinical research and drug discovery.

⁵ McKinsey (January 2026), "Transforming Europe: Bold Moves to Lift a Continent". Novo Nordisk: >\$8.2 Bn R&D in 2024, using digital twins and AI for drug discovery.

⁶ McKinsey (December 2025), "Accelerating Europe's AI Adoption". The report observes that AI productivity gains are concentrated in "lighthouse" companies, with gains reaching 40% labor productivity and 50% time-to-market reduction.

⁷ Segler Consulting (June 2025), "Europe's AI Gambit". Total EU public capacity is estimated at ~57,000 accelerators in 2025, an order of magnitude below the infrastructure of a single US hyperscaler.

⁸ Mistral AI (September 2025), Series C: €1.7 Bn raised, €11.7 Bn valuation. ASML lead investor (€1.3 Bn for ~11%). Other investors: DST Global, a16z, Bpifrance, General Catalyst, Index Ventures, Lightspeed, Nvidia.

⁹ Introl Blog (December 2025), "France's AI Sovereignty Push". Mistral Compute: 18,000 Grace Blackwell GPUs, 40 MW data center in Essonne (hosted by Scaleway/Eclairion), powered by nuclear energy. Launch planned 2026.

¹⁰ Mistral AI / Wikipedia FR (February 2026). Borlänge data center (Sweden): €1.2 Bn, planned for 2027, via EcoDataCenter (renewable energy). Acquisition of Koyeb (February 17, 2026) to strengthen Mistral Compute's serverless offering.

¹¹ Dealroom (2025), cited in FinTech Weekly. AI funding in Europe +55% in Q1 2025, 12 new unicorns in H1 2025. France is the top EU destination for foreign AI investment since 2020.

¹² Ministry of the Armed Forces (February 2025), launch of GenIAI.intradef. Secure conversational agent based on Mistral AI, deployed on sovereign infrastructure.

¹³ Martens, B. (2024), Working Paper 18/2024, Bruegel, op. cit. AI personnel costs (salaries + equity) are often the most significant component of model development, and US companies offer structurally superior packages.

¹⁴ Martens, B. (2025), "The European Union Needs More Than the Digital Omnibus to Make Digital Services Competitive", Bruegel. The AI Omnibus, intended to accelerate regulatory changes, risks taking at least one year to be adopted, plus one year for supplementary guidelines.

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